

ASSIGNMENT NO.5: Aggregate Functions and Data Analysis

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Questions:

Q1) Create a MySQL database named CollegeDB and a table named StudentResult.

The table should contain the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
FeesPaid	DECIMAL(10,2)	

Insert at least 10 records into the table.

QUERY:

```
1 CREATE DATABASE SY_CollegeDB;
2 USE SY_CollegeDB;
3
4 CREATE TABLE StudentResult (
5     StudentID INT PRIMARY KEY,
6     StudentName VARCHAR(50),
7     Course VARCHAR(30),
8     City VARCHAR(30),
9     Marks INT,
10    FeesPaid DECIMAL(10,2)
11 );
12
13 INSERT INTO StudentResult VALUES
14 (1, 'Aarav Shah', 'BSc IT', 'Mumbai', 85, 75000),
15 (2, 'Meera Joshi', 'BCA', 'Pune', 78, 70000),
16 (3, 'Rohan Patil', 'BSc CS', 'Nashik', 88, 72000),
17 (4, 'Sneha Desai', 'BCA', 'Mumbai', 92, 70000),
18 (5, 'Amit Kumar', 'BSc IT', 'Delhi', 75, 75000),
19 (6, 'Pooja Singh', 'BSc CS', 'Pune', 81, 72000),
20 (7, 'Vikram Rao', 'BCA', 'Bengaluru', 89, 70000),
21 (8, 'Neha Sharma', 'BSc IT', 'Nashik', 94, 75000),
22 (9, 'Rahul Verma', 'BSc CS', 'Mumbai', 73, 72000),
23 (10, 'Kavita Iyer', 'BCA', 'Chennai', 87, 70000);
24
25 SELECT * FROM StudentResult;
```

OUTPUT:

		StudentID	StudentName	Course	City	Marks	FeesPaid
☐	Edit Copy Delete	1	Aarav Shah	BSc IT	Mumbai	85	75000.00
☐	Edit Copy Delete	2	Meera Joshi	BCA	Pune	78	70000.00
☐	Edit Copy Delete	3	Rohan Patil	BSc CS	Nashik	88	72000.00
☐	Edit Copy Delete	4	Sneha Desai	BCA	Mumbai	92	70000.00
☐	Edit Copy Delete	5	Amit Kumar	BSc IT	Delhi	75	75000.00
☐	Edit Copy Delete	6	Pooja Singh	BSc CS	Pune	81	72000.00
☐	Edit Copy Delete	7	Vikram Rao	BCA	Bengaluru	89	70000.00
☐	Edit Copy Delete	8	Neha Sharma	BSc IT	Nashik	94	75000.00
☐	Edit Copy Delete	9	Rahul Verma	BSc CS	Mumbai	73	72000.00
☐	Edit Copy Delete	10	Kavita Iyer	BCA	Chennai	87	70000.00

Q2) Write MySQL queries to find the following from the StudentResult table:

Requirement
Total number of students
Total fees paid
Average marks
Lowest marks
Highest marks

QUERY:

```
1 SELECT
2     COUNT(*) AS TotalStudents,
3     SUM(FeesPaid) AS TotalFeesPaid,
4     AVG(Marks) AS AverageMarks,
5     MIN(Marks) AS LowestMarks,
6     MAX(Marks) AS HighestMarks
7 FROM StudentResult;
```

OUTPUT:

TotalStudents	TotalFeesPaid	AverageMarks	LowestMarks	HighestMarks
10	721000.00	84.2000	73	94

Q3) Write a MySQL query to count the number of students in each course.

The output should contain:

Column
Course
TotalStudents

QUERY:

```
1 SELECT Course, COUNT(*) AS TotalStudents
2 FROM StudentResult
3 GROUP BY Course;
```

OUTPUT:

Course	TotalStudents
BCA	4
BSc CS	3
BSc IT	3

Q4) Write a MySQL query to find the average marks of students course-wise.

The output should contain:

Column
Course
AverageMarks

QUERY:

```
1 SELECT Course, AVG(Marks) AS AverageMarks
2 FROM StudentResult
3 GROUP BY Course;
```

OUTPUT:

	Course	AverageMarks
☐  Edit  Copy  Delete	BCA	86.5000
☐  Edit  Copy  Delete	BSc CS	80.6667
☐  Edit  Copy  Delete	BSc IT	84.6667

Q5) Write a MySQL query to display only those courses where the number of students is greater than 2.

Use:

Clause
GROUP BY
HAVING

QUERY:

```
1 SELECT Course, COUNT(*) AS TotalStudents
2 FROM StudentResult
3 GROUP BY Course
4 HAVING COUNT(*) > 2;
```

OUTPUT:

Course	TotalStudents
BCA	4
BSc CS	3
BSc IT	3

Q6) Write a MySQL query to analyze course performance.

The output should display:

Column
Course
Column
TotalStudents
AverageMarks
LowestMarks
HighestMarks

Sort the result by AverageMarks in descending order.

QUERY:

```
1 SELECT Course,
2     COUNT(*) AS TotalStudents,
3     AVG(Marks) AS AverageMarks,
4     MIN(Marks) AS LowestMarks,
5     MAX(Marks) AS HighestMarks
6 FROM StudentResult
7 GROUP BY Course
8 ORDER BY AverageMarks DESC;
```

OUTPUT:

Course	TotalStudents	AverageMarks ▾ 1	LowestMarks	HighestMarks
BCA	4	86.5000	78	92
BSc IT	3	84.6667	75	94
BSc CS	3	80.6667	73	88